

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 2 – Technical Presentation

Course Code:	DSA03	Course Title:	Natural Language Processing
CO Assessed:	CO1 – Imitation (L1)	Assessment:	Assessment Tool 2 – Technical Presentation
Weightage:	30% of CIA		
CO1 Statement:	Analyse NLP models, regular expression patterns, finite state automata, morphological processing techniques and to interpret language processing. (BL-3)		
Student Name:	S.Dhanashree,T S Sai Lahari	Reg. No.:	192325082,19232537
Section / Batch:	AIML	Date:	

Code of Conduct

I _S.Dhanashree(192325082)_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: S.Dhanashree

Instructions

- Suggested Team Member Roles (3 Members)
- Member 1: Theory, Concepts, and Literature Review
- Member 2: Algorithm Design, Models, and Technical Analysis
- Member 3: Demonstration, Case Study, Applications, and Conclusion
- GPS photo must submit.
- Duration: 30 minutes.

Section / Topic	Focus	Q Nos
Evolution of Natural Language Processing: From Rule-Based Systems to Modern NLP	NLP Models, Language Processing Evolution	TP-01
Comparative Study of NLP Models and Algorithms	Model Analysis, Algorithm Comparison	TP-02
Regular Expressions for Real-World Text Processing	Pattern Matching, Information Extraction	TP-03
Designing Efficient Regular Expression Patterns for Information Extraction	Text Mining, Data Extraction Techniques	TP-04
Finite State Automata in Natural Language Processing	State Transitions, Language Recognition	TP-05
Building a Finite State Automaton for Token Recognition	Lexical Analysis, Token Validation	TP-06
Morphology and Word Formation in Natural Languages	Morpheme Analysis, Word Structure Processing	TP-07
Inflectional Morphology: Understanding Grammatical Variations	Grammatical Analysis, Word Normalization	TP-08
Derivational Morphology and Vocabulary Expansion	Word Generation, Semantic Transformation	TP-09
Comparative Analysis of Inflectional and Derivational Morphology	Morphological Classification, Language Analysis	TP-10

Finite-State Morphological Parsing Techniques	Morphological Parsing, Finite-State Processing	TP-11
Morphological Analysis for Indian Languages	Language-Specific Morphology, NLP Challenges	TP-12
Porter Stemmer: Design, Working Mechanism, and Applications	Stemming Algorithms, Text Preprocessing	TP-13
Stemming vs Morphological Parsing: A Comparative Study	Word Reduction, Morphological Processing	TP-14
Integrated NLP Pipeline: From Regular Expressions to Morphological Processing	End-to-End Language Processing Pipeline	TP-15

Annexure A – Technical Presentation

Team No.	Technical Presentation Title	Description
1	Evolution of Natural Language Processing: From Rule-Based Systems to Modern NLP	Analyze the historical development of NLP models, key milestones, and how language processing has evolved from handcrafted rules to intelligent systems.
2	Comparative Study of NLP Models and Algorithms	Explore different NLP models and algorithms used for language processing, comparing their working principles, strengths, and limitations.
3	Regular Expressions for Real-World Text Processing	Demonstrate how regular expressions are used for data validation, text extraction, log analysis, and information retrieval in practical applications.
4	Designing Efficient Regular Expression Patterns for Information Extraction	Analyze complex regular expression patterns used to extract emails, phone numbers, URLs, dates, and structured information from unstructured text.
5	Finite State Automata in Natural Language Processing	Explain the concepts of deterministic and non-deterministic finite automata and their role in lexical analysis and language recognition.
6	Building a Finite State Automaton for Token Recognition	Design and demonstrate a finite state automaton that recognizes tokens such as identifiers, keywords, and numbers in textual data.
7	Morphology and Word Formation in Natural Languages	Investigate how words are structured and formed using morphemes, and analyze the importance of morphology in NLP systems.
8	Inflectional Morphology: Understanding Grammatical Variations	Examine how inflectional changes affect word forms based on tense, number, gender, and case across different languages.
9	Derivational Morphology and Vocabulary Expansion	Analyze derivational processes that create new words and study their impact on semantic interpretation and language understanding.
10	Comparative Analysis of Inflectional and Derivational Morphology	Compare the characteristics, applications, and challenges of inflectional and derivational morphology in NLP tasks.
11	Finite-State Morphological Parsing Techniques	Explore finite-state transducers and parsing techniques used to analyze and generate morphological structures in natural languages.
12	Morphological Analysis for Indian Languages	Study the challenges and solutions involved in performing morphological processing for morphologically rich Indian languages such as Tamil, Hindi, and Telugu.
13	Porter Stemmer: Design, Working Mechanism, and Applications	Analyze the Porter Stemming algorithm, its rule-based approach, implementation steps, and role in information retrieval systems.
14	Stemming vs Morphological Parsing: A Comparative Study	Compare stemming and morphological parsing techniques based on accuracy, computational complexity, and application scenarios.
15	Integrated NLP Pipeline: From Regular Expressions to Morphological Processing	Demonstrate a complete NLP preprocessing pipeline incorporating regular expressions, finite state automata, stemming, and morphological analysis for text processing.

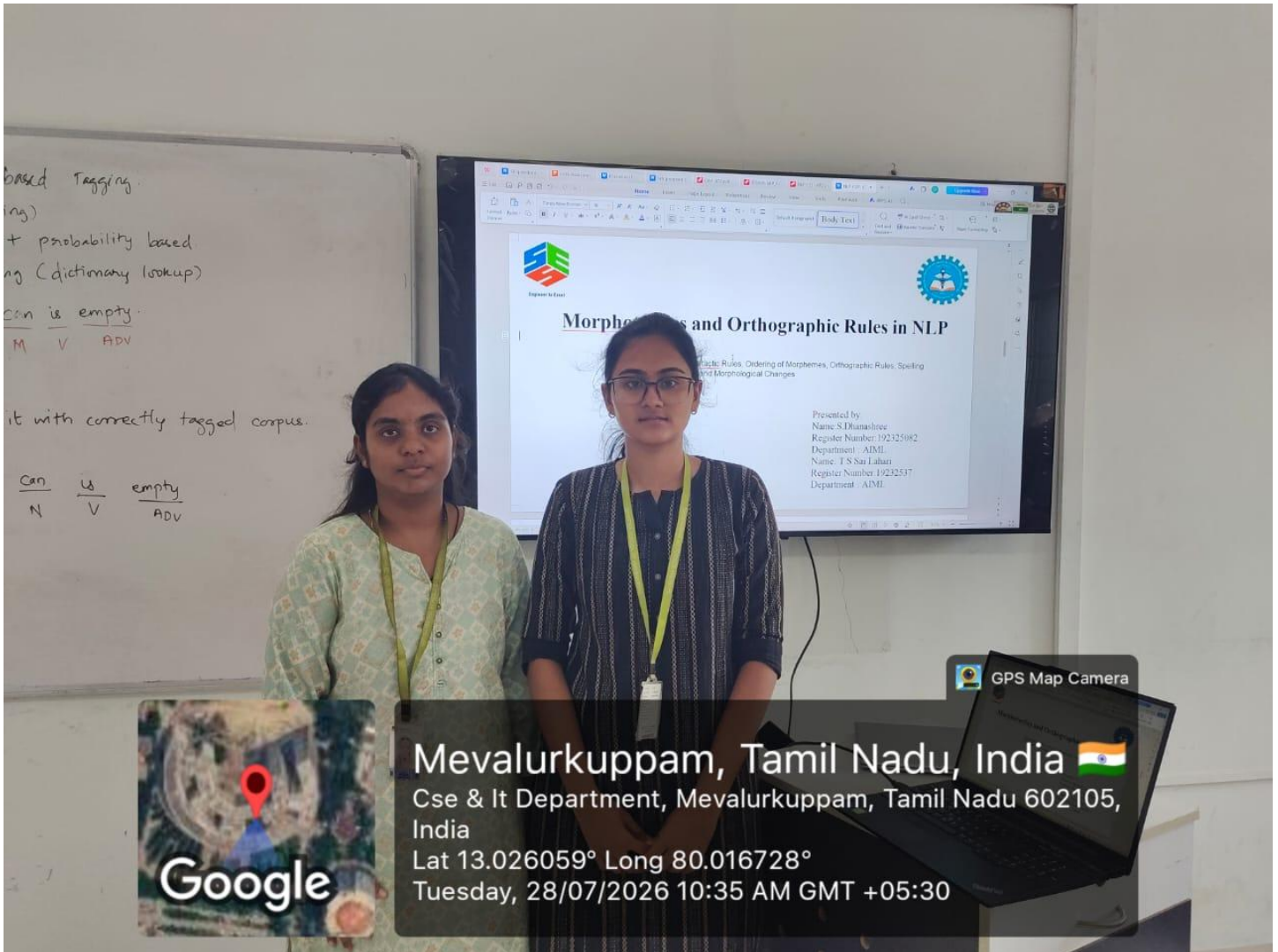
Rubrics for Calculation

Technical Presentation					
Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Technical Content Accuracy	30	Highly accurate, in-depth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well-analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness

Team No. / Criteria Level	Technical Content Accuracy	Problem Identification	Use of Tools	Communication	Professionalism	Total %
1	30	25	20	12	10	97

Faculty In-charge	Course Coordinator	Program Director
Kumaragurubaran T		
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Geo Tag Photo:





Morphotactics and Orthographic Rules in NLP

Morphotactic Rules, Ordering of Morphemes, Orthographic Rules, Spelling Rules, and Morphological Changes

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Introduction

- Morphology is the branch of linguistics that studies the structure of words.
- It explains how words are formed using morphemes, the smallest meaningful units.
- It plays an important role in Natural Language Processing (NLP).

- Applications include:

- Machine Translation
- Spell Checking
- Information Retrieval
- Speech Recognition
- Text Processing

Word	Morphemes
Unhappiness	Un + Happy + ness
Played	Play + ed
Books	Book + s
Running	Run + ing



Morphotactics



Morphotactics is the study of how morphemes are arranged to create valid words.

It defines:

Which morphemes can combine

The correct order of morphemes

Restrictions on combinations

Example

Correct:

play + er = player

teach + er = teacher

quick + ly = quickly

Incorrect:

er + teach

ly + quick

ness + happy

Morphotactics ensures words follow grammatical rules.



Morphotactic Rules



- These rules define the legal arrangement of morphemes.

General Structure

Prefix + Root + Suffix

Examples

Prefix	Root	Suffix	Word
un	happy	ness	unhappiness
re	write	ing	rewriting
dis	connect	ion	disconnection

Important Rules

- Root is mandatory.
- Prefix comes before root.
- Suffix comes after root.
- Not every prefix works with every root.



Ordering of Morphemes



Morphemes appear in a fixed order.

Standard Order

Prefix → Root → Derivational Suffix → Inflectional Suffix

Example 1

un + believe + able

↓

Unbelievable

Example 2

play + er + s

↓

players

Example 3

modern + ize + ation

↓

Modernization

Changing the order creates invalid words.

Example:

play + s + er ✗



Valid and Invalid Morpheme Combinations

Valid Combinations

Combination	Result
read + ing	reading
write + er	writer
kind + ness	kindness
help + ful	helpful
move + ment	movement

Invalid Combinations

Combination	Reason
ing + read	Wrong order
ness + happy	Prefix missing
ful + help	Incorrect arrangement
s + book	Wrong placement

Morphotactic rules reject these combinations.



Orthographic Rules



- Orthographic rules explain how spelling changes when morphemes combine.
- These changes occur because English spelling follows pronunciation patterns.
- Examples include:
 - Changing y to i
 - Doubling consonants
 - Dropping silent e
 - Adding suffixes
- These rules improve readability and pronunciation.

Spelling Rules

Common Spelling Rules

Rule 1

Change y → ies
city → cities
baby → babies
lady → ladies

Rule 2

If vowel + y
Simply add s
boy → boys
toy → toys
day → days

Rule 3

Drop silent e
make → making
write → writing
hope → hoping

Rule 4

Double final
consonant
run → running
sit → sitting
stop → stopped



Morphological Changes During Word Formation



Morphological Changes

When suffixes are added, the spelling may change.

Types

1. Addition

book → books

cat → cats

2. Replacement

city → cities

fly → flies

3. Deletion

make → making

arrive → arriving

4. Doubling

run → running

begin → beginning

These changes preserve pronunciation.



Examples



Examples of Word Formation

Root Word	Rule Applied	New Word
city	y → ies	cities
fly	y → ies	flies
make	Drop e	making
run	Double consonant	running
stop	Double consonant	stopped
teach	Add suffix	teacher
happy	Add ness	happiness
care	Add ful	careful



Applications in NLP



Importance in Natural Language Processing

Morphotactics and orthographic rules help computers understand language.

Applications

- Lemmatization
- Stemming
- Spell checking
- Machine Translation
- Search Engines
- Speech Recognition
- OCR correction
- Grammar Checking
- Text preprocessing

Example

Input:

cities



Morphological Analyzer



city + plural



Python Program – Orthographic Rule (y → ies)



Python program to apply the spelling rule:

- If a word ends with 'y' and the previous letter is a consonant, replace 'y' with 'ies'.
- Otherwise, simply add 's'.

Sample Output

Input 1

Enter a word: city

Output

Plural form: cities

Code

```
word = input("Enter a word: ")
```

```
vowels = "aeiou"
```

```
if word.endswith("y") and word[-2].lower() not in vowels:
```

```
    plural = word[:-1] + "ies"
```

```
else:
```

```
    plural = word + "s"
```

```
print("Plural form:", plural)
```

Input 2

Enter a word: boy

Output

Plural form: boys



Thank you